Cairo University – Faculty of Engineering
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Machine Learning For Industrial Engineering

Project Assignment
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Executive Summary

This report compares three supervised machine learning models for predicting machine failure: Logistic Regression, Support Vector Machine (SVM), and Neural Network. Each model was trained using the same train-test split and evaluated using confusion matrix, accuracy, precision, recall, specificity, F1-score, ROC curve, and AUC. The calculations are highlighted in the metrics section to make the model comparison clear and easy to verify.

Main finding: The Neural Network achieved the strongest overall discrimination performance, with the highest AUC and best recall/F1-score balance among the three models.

Dataset Overview

The dataset used is the AI4I 2020 predictive maintenance dataset. It contains 10,000 observations and includes rotational speed, torque, and a binary machine failure label. The UDI column was excluded because it is only an identifier and does not carry predictive information.

Item	Value
Records	10,000
Target variable	Machine failure
Feature 1	Rotational speed [rpm]
Feature 2	Torque [Nm]
Dropped column	UDI

Methodology

The dataset was split into 70% training data and 30% testing data using stratification so that the failure/non-failure distribution remained consistent. Numerical features were scaled using StandardScaler before training each model.

The trained models were: Logistic Regression, SVM with an RBF kernel, and a feed-forward Neural Network using MLPClassifier with hidden layers of 50 and 25 neurons.

Calculation Formulae

The following highlighted formulae were used to calculate the model performance metrics from each confusion matrix.

Accuracy: (TP + TN) / (TP + TN + FP + FN) - overall proportion of correct predictions

Precision: TP / (TP + FP) - how many predicted failures were actual failures

Recall: TP / (TP + FN) - how many actual failures were detected

Specificity: TN / (TN + FP) - how many normal cases were correctly identified

F1-Score: 2 x Precision x Recall / (Precision + Recall) - balance between precision and recall

AUC: Area under the ROC curve - overall ranking ability of the model

Model Evaluation Results

Model	Accuracy	Precision	Recall	Specificity	F1-Score	AUC
Logistic Regression	0.969	**0.917**	0.108	**1.000**	0.193	0.797
SVM	0.970	0.800	0.157	0.999	0.262	0.480
Neural Network	**0.971**	0.792	**0.186**	0.998	**0.302**	**0.877**

Table 1: Highlighted performance calculations for all trained models.

Model	TN	FP	FN	TP
Logistic Regression	**2897**	**1**	91	11
SVM	2894	4	86	16
Neural Network	2893	5	**83**	**19**

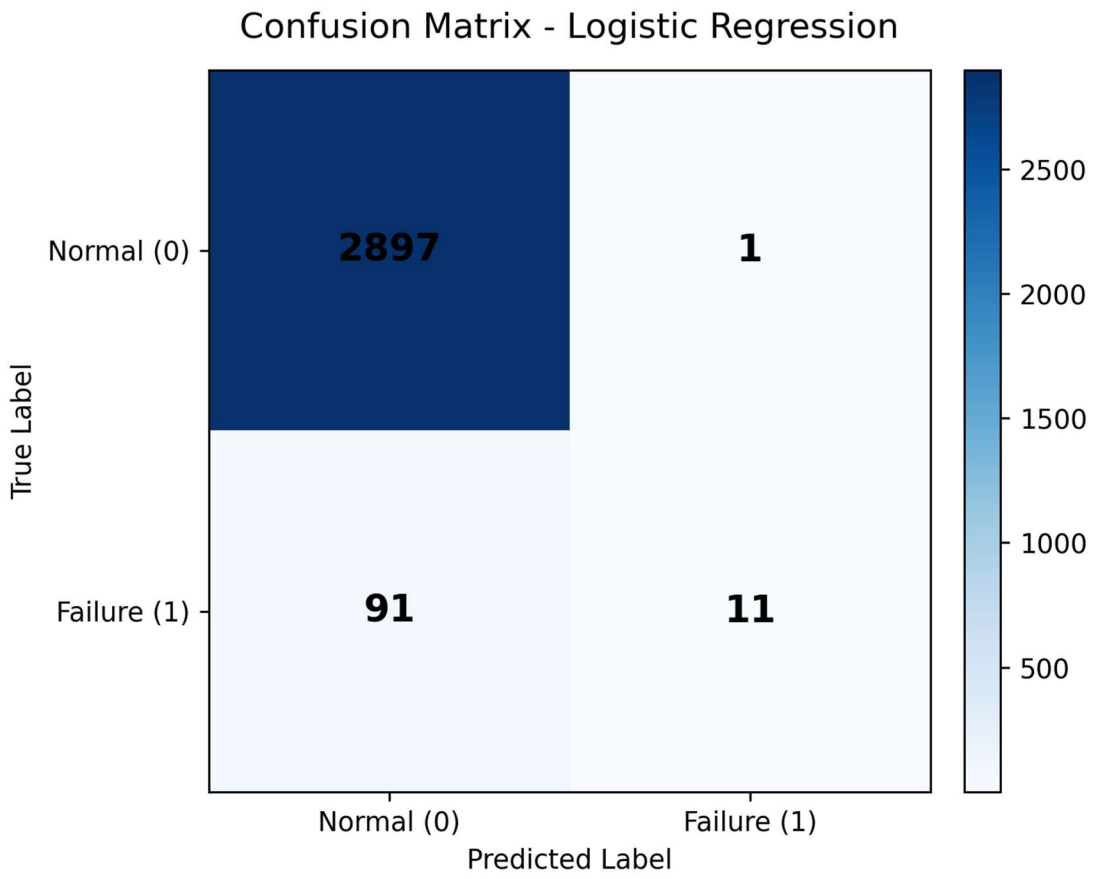
Table 2: Highlighted confusion-matrix counts used in the metric calculations.

The highlighted cells show the calculated metric values and confusion-matrix counts. The darker highlighted cells mark the strongest result for each metric. Neural Network achieved the highest Accuracy, Recall, F1-Score, AUC, and TP count, while Logistic Regression achieved the highest Precision and Specificity.

Graphical Analysis

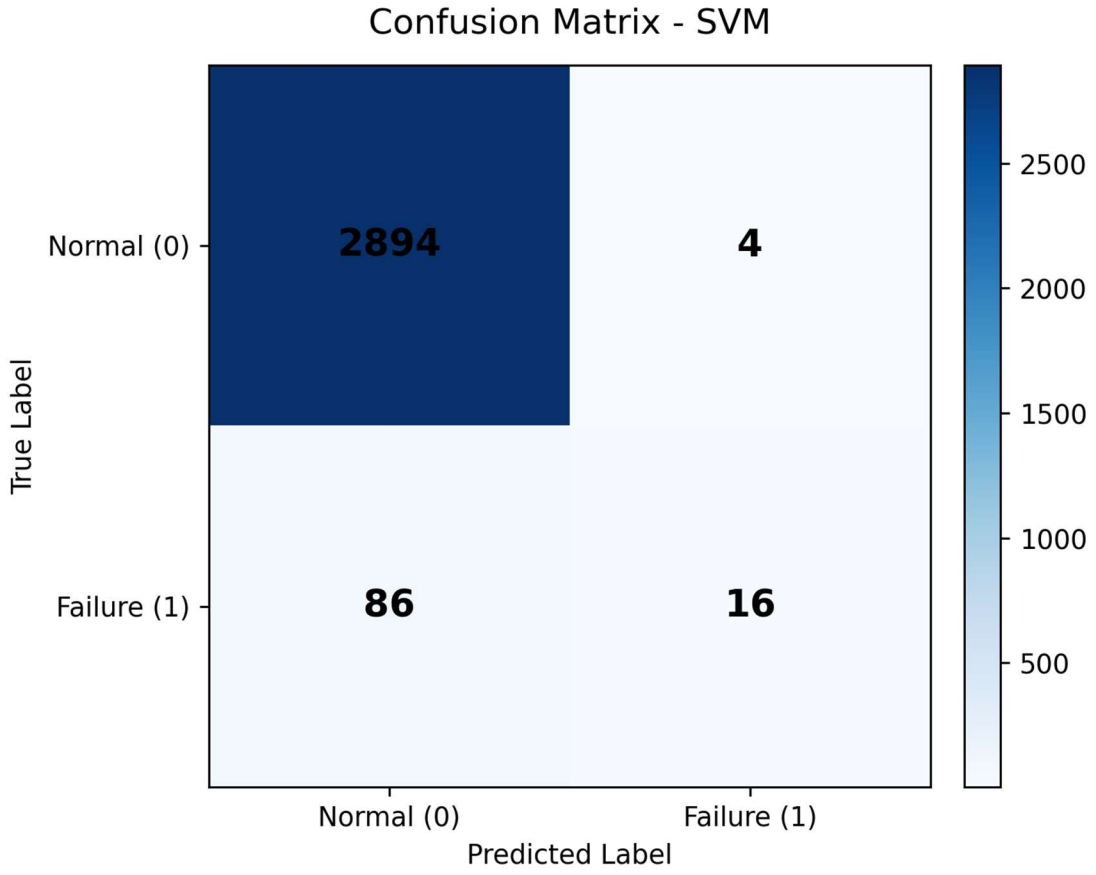
Confusion Matrices

Figure 1: Confusion Matrix - Logistic Regression



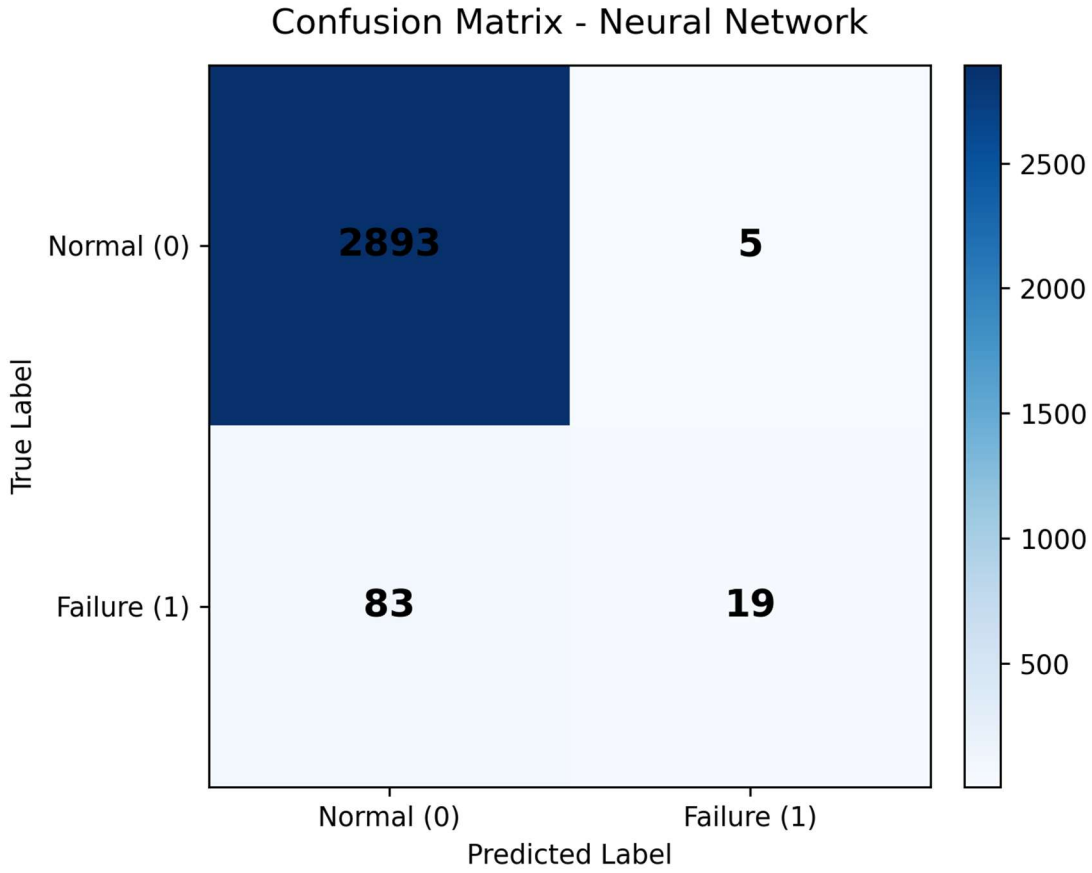
Logistic Regression produced very few false positives, which explains its strong precision and specificity. However, it detected only a small portion of actual failure cases, resulting in low recall.

Figure 2: Confusion Matrix - SVM



The SVM produced slightly more true positives than Logistic Regression but also had more false positives. Its recall improved slightly, but the ROC/AUC performance was weak compared with the Neural Network.

Figure 3: Confusion Matrix - Neural Network



The Neural Network achieved the highest number of true positives and the highest recall, making it the best model for detecting failures among the three tested models.

ROC and Overall Model Comparison

Figure 4: ROC Curve Comparison

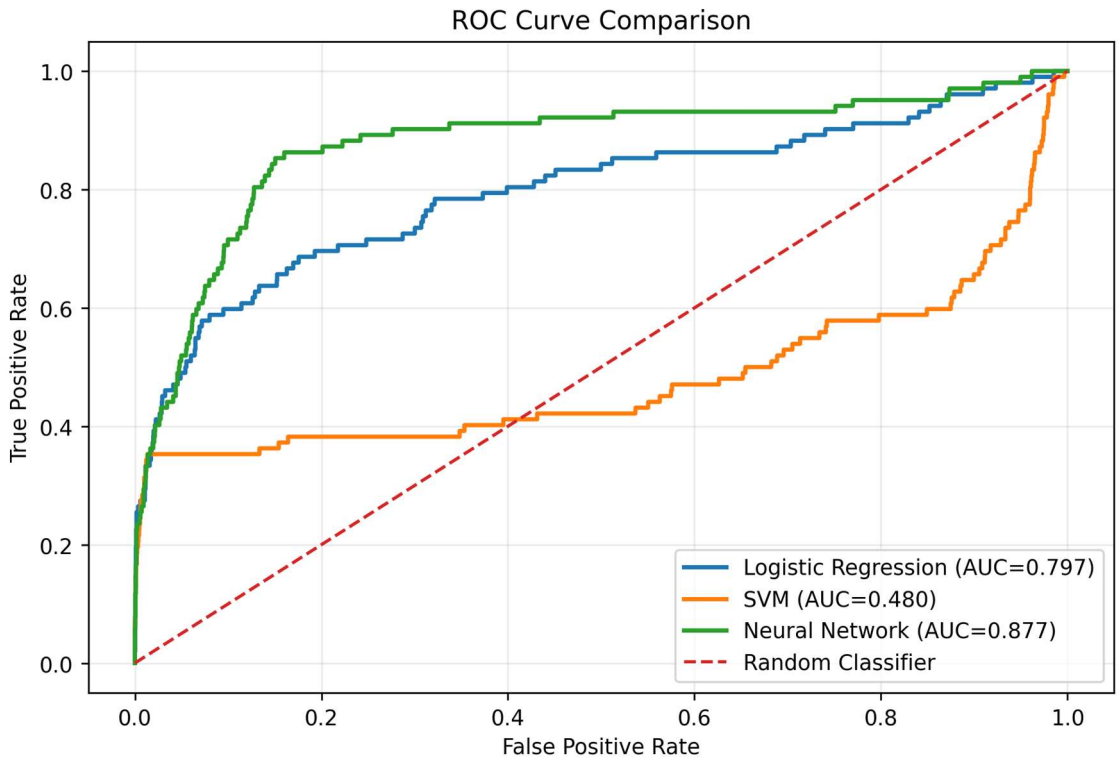
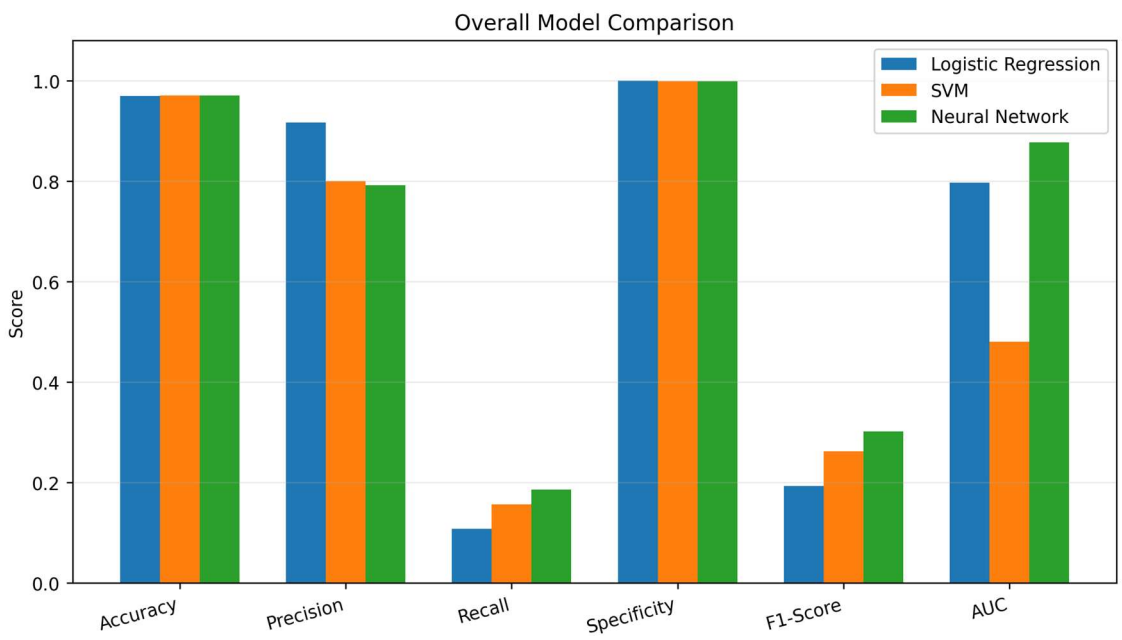


Figure 5: Overall Model Performance Comparison



Metric Comparison Graphs

Figure 6: Accuracy Comparison

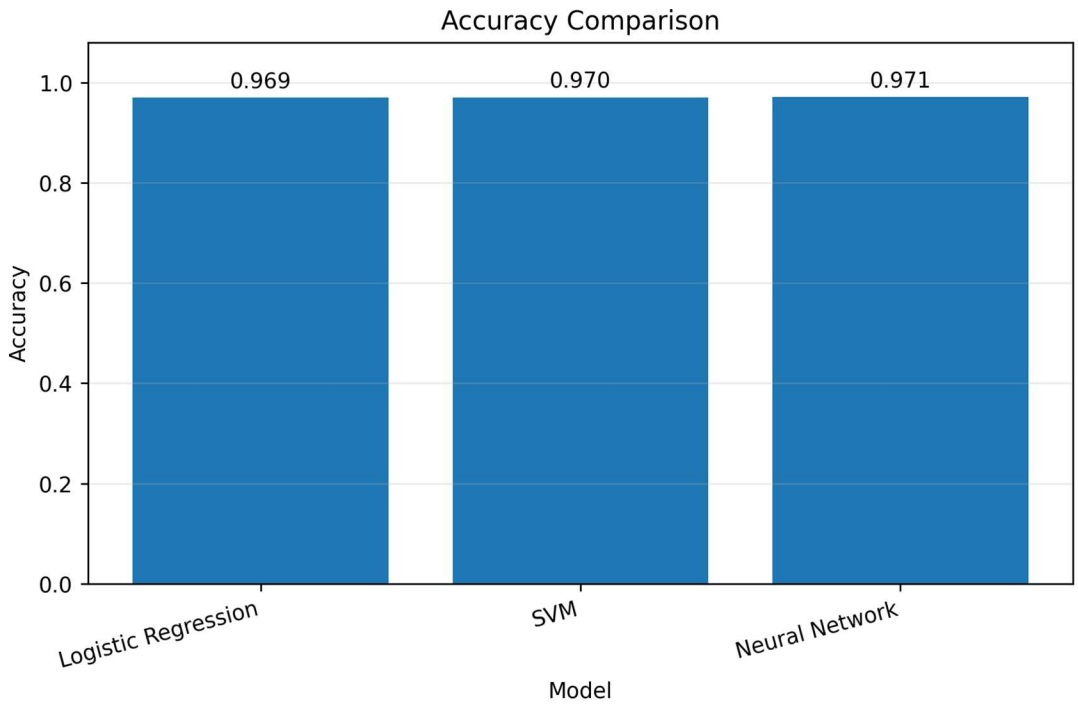


Figure 7: Precision Comparison

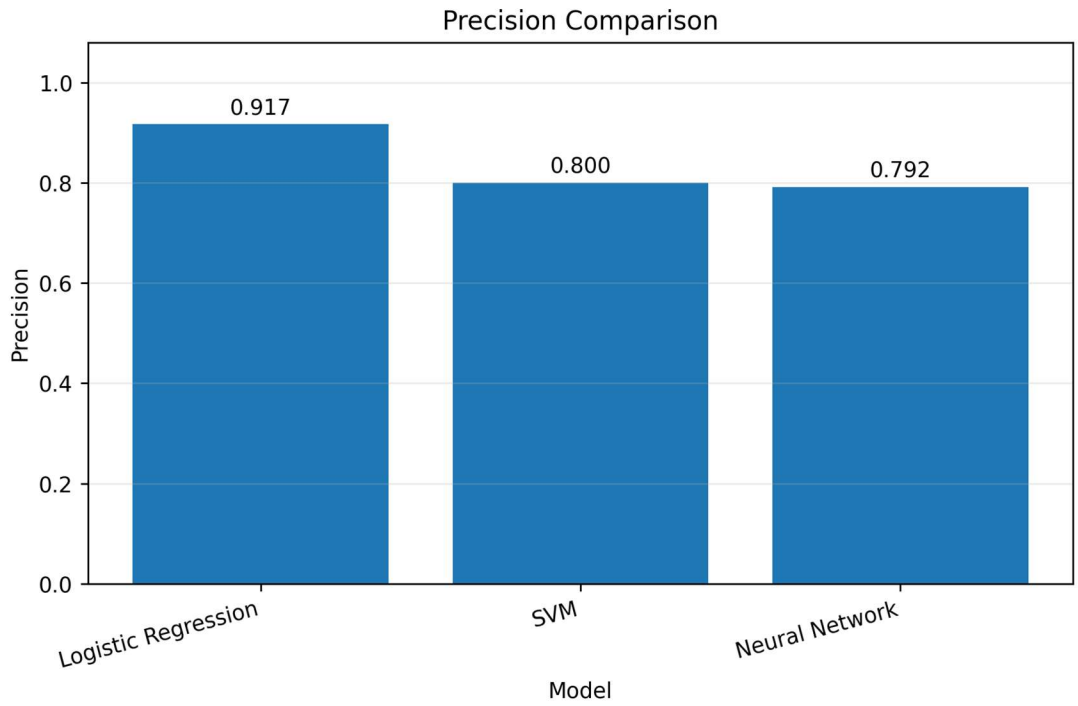


Figure 8: Recall Comparison

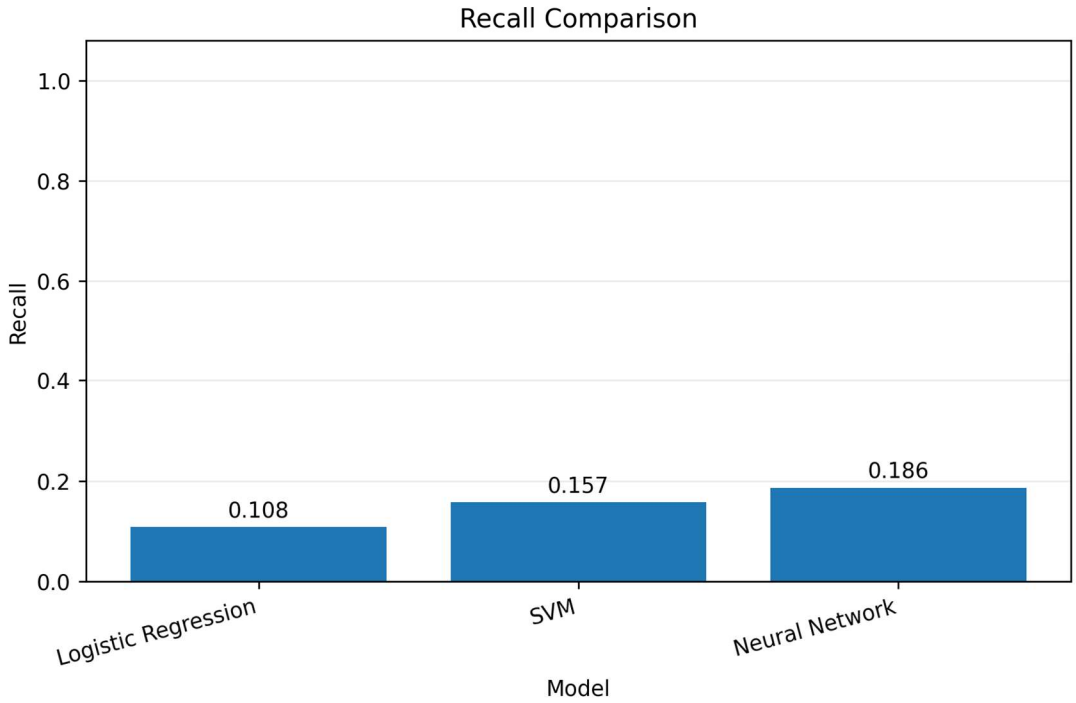


Figure 9: Specificity Comparison

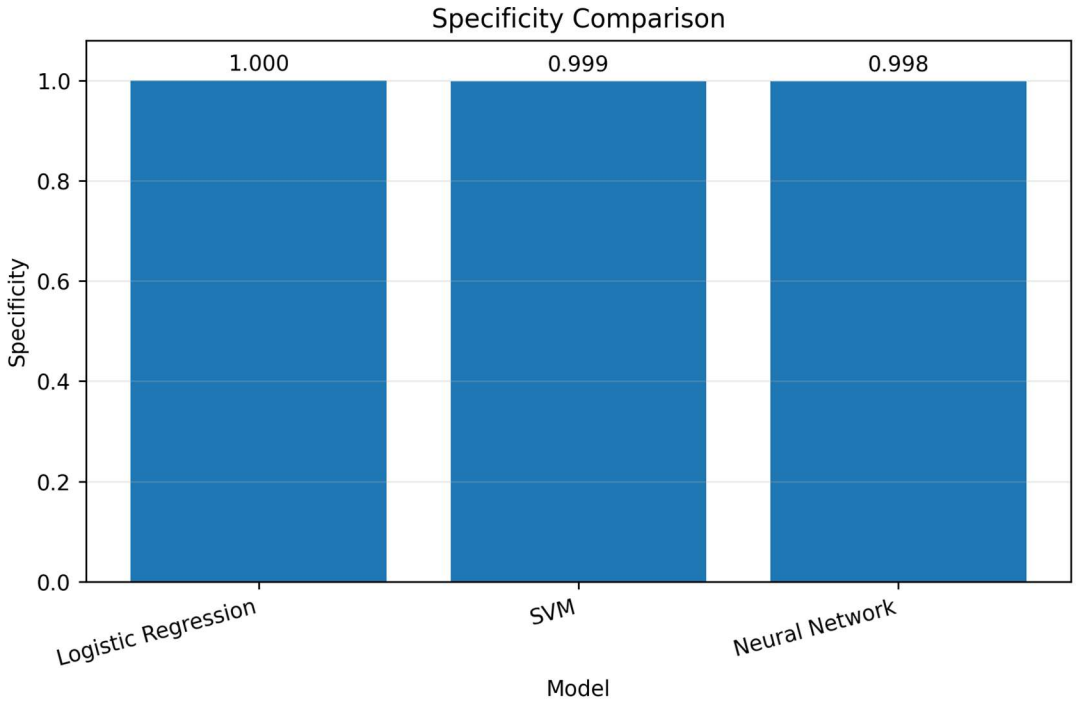


Figure 10: F1-Score Comparison

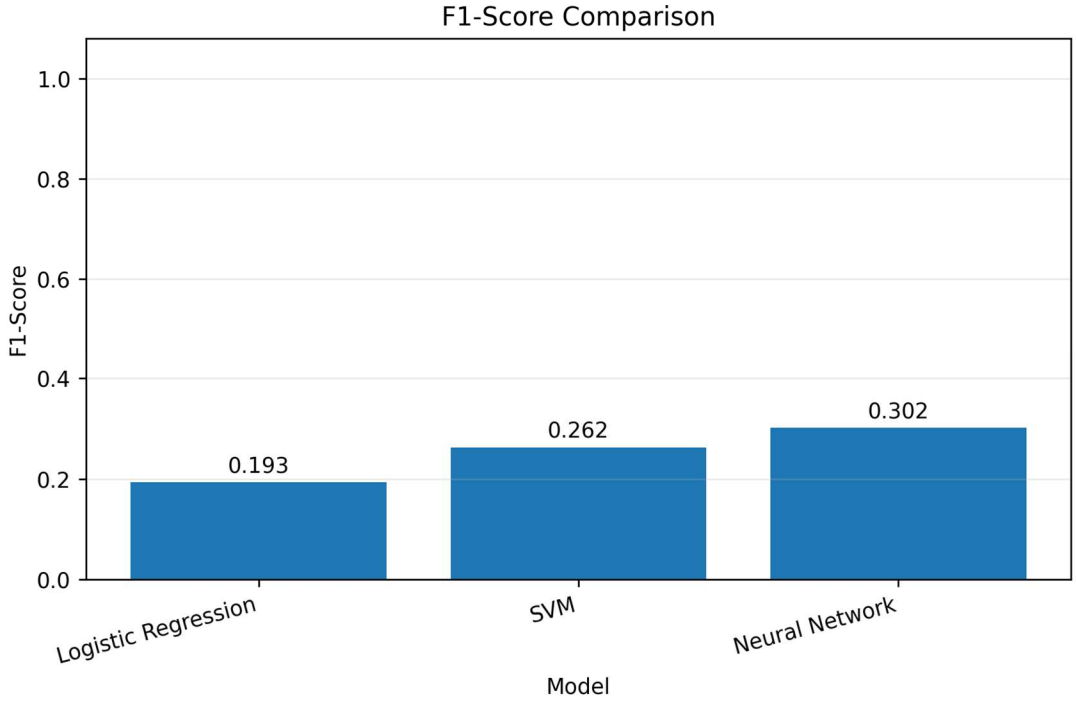
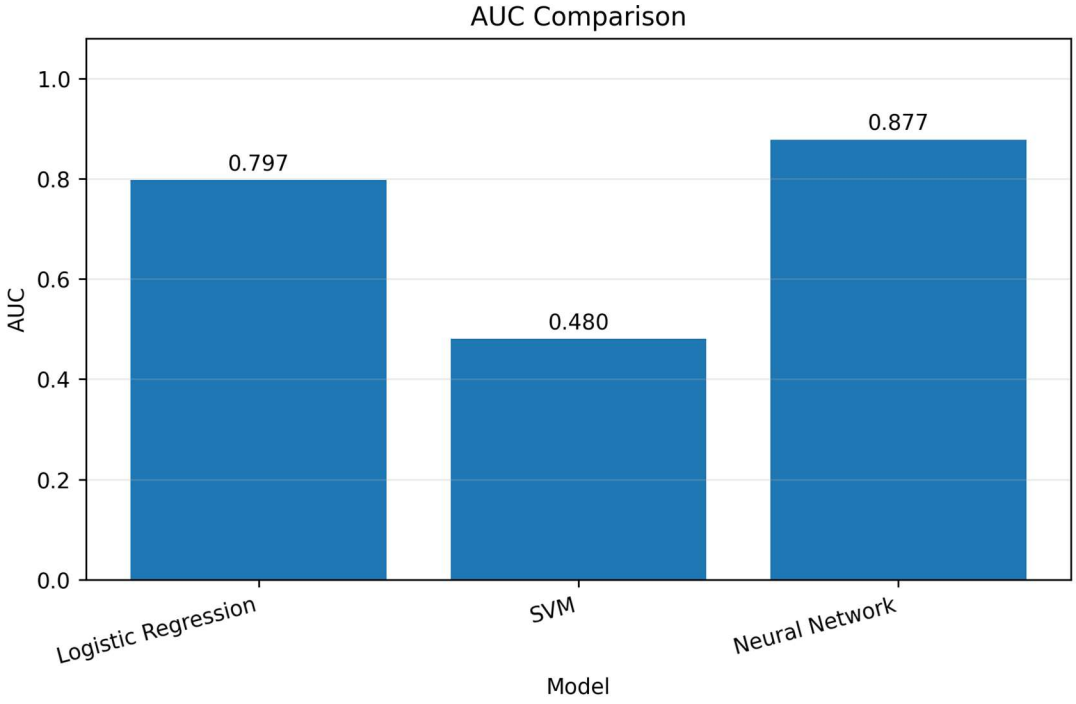


Figure 11: AUC Comparison



Discussion and Final Recommendation

The three models produced very high accuracy values, but this is strongly affected by class imbalance because most records represent non-failure cases. For a predictive maintenance problem, recall, F1-score, and AUC are more important than accuracy alone because the main objective is to detect failure cases before they happen.

Logistic Regression achieved the highest precision and specificity, meaning it was conservative in predicting failures. However, its recall was the lowest, so it missed many actual failures.

The SVM produced slightly better recall than Logistic Regression, but its AUC was the weakest. This indicates that, in this configuration, the SVM was not the best discriminator between failure and non-failure cases.

The Neural Network achieved the best overall balance. It produced the highest recall, F1-score, AUC, and true positive count. Therefore, the Neural Network is the recommended model for this dataset.

Recommended Model	Neural Network
Reason	**Highest AUC, Recall, F1-Score, and True Positive count**

Conclusion

The final comparison confirms that the Neural Network is the strongest model for predicting machine failure in this experiment. While Logistic Regression and SVM achieve similar accuracy, the Neural Network provides better failure detection capability, which is more valuable for industrial predictive maintenance.

Future improvement should focus on handling class imbalance through class weights, SMOTE, or other resampling methods. Additional sensor features would also likely improve recall and reduce missed failures.